

Q1

Imagine a flat playground spinner. The distance between the center and the edge of the spinner is $r = 3\text{ m}$. It is a perfectly flat, solid circular platform that spins horizontally.

- (a) The spinner rotates through an angle of $\theta = 4\text{ rad}$. Calculate the actual distance you traveled if you sit at the edge.
- (b) If it takes $t = 2\text{ s}$ to complete this rotation, calculate the angular velocity (ω) of the spinner.
- (c) Calculate your linear speed.



Q2

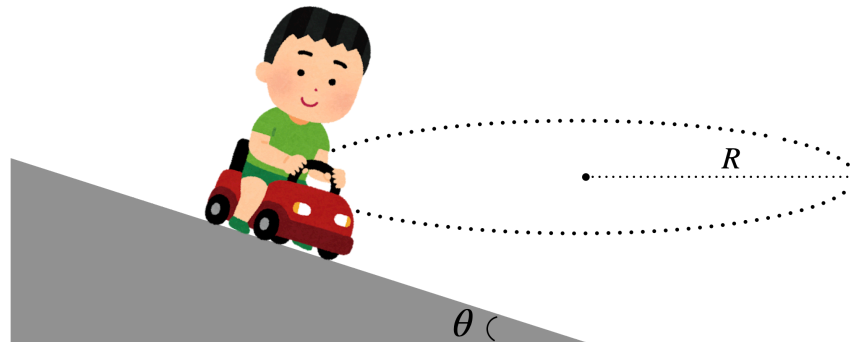
Imagine you place a small coin on the very edge of the same spinner from Q1.

- (a) Calculate the centripetal acceleration of the coin while it is turning on the spinner.
- (b) Which direction does this centripetal acceleration point? Depict circle as the top view of the spinner, and denote the acceleration arrow.
- (c) Suddenly, the coin slips! Which direction does the coin fly in the horizontal direction? Explain using Newton's 1st Law.

Q3

Imagine a 1200 kg car turning on a smooth, frictionless road that is banked (tilted) at an angle of $\theta = 37^\circ$. The car is making a HORIZONTAL circle.

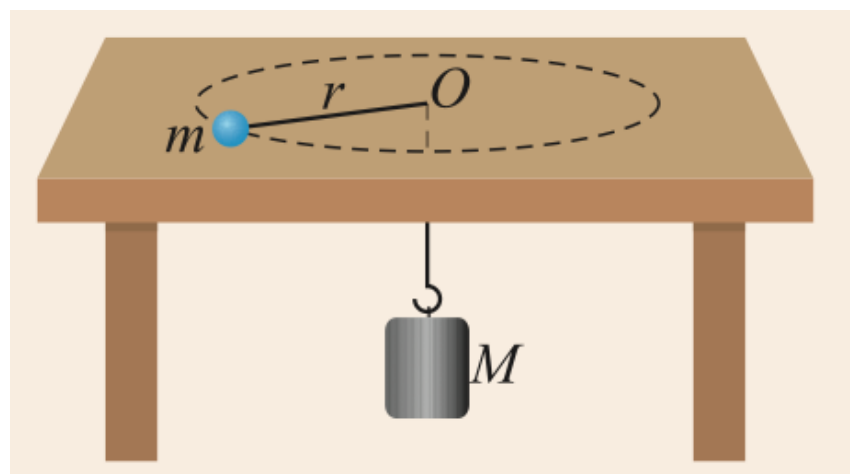
- (a) Calculate the exact value of the normal force.
- (b) Calculate the centripetal force (F_c).
- (c) Calculate the centripetal acceleration (a_c) of the car.



Q4

A small blue ball with mass $m = 5 \text{ kg}$ is on a frictionless horizontal table. It is tied to a string that passes through a hole in the center to a hanging weight with mass $M = 4 \text{ kg}$. The blue ball is swinging in a horizontal circle with a radius of $r = 2 \text{ m}$.

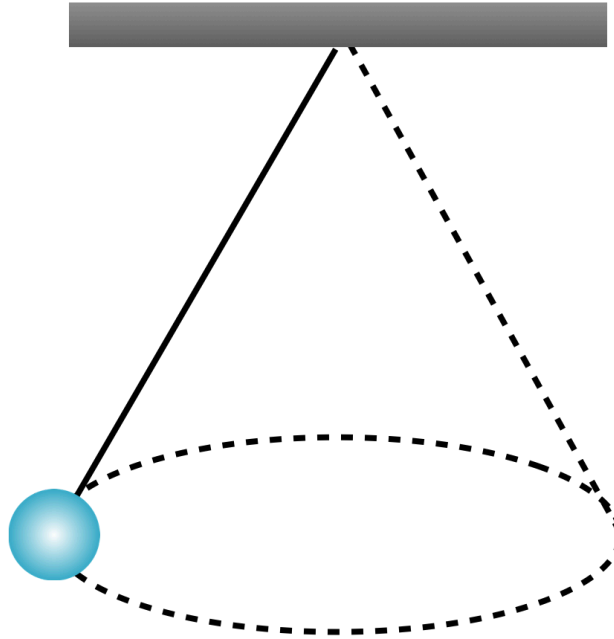
- (a) Calculate the Tension in the string.
- (b) Calculate the centripetal acceleration (a_c) of the blue ball.
- (c) Calculate the blue ball's tangential speed.



Q5

A ball with a mass of $m = 1.2 \text{ kg}$ is flying in a horizontal circle, suspended by a string of length $L = 1.5 \text{ m}$. The string makes an angle of $\theta = 53^\circ$ with the vertical straight line.

- (a) Calculate the exact value of the tension.
- (b) Calculate the radius R of the horizontal circle and calculate the centripetal force.
- (c) Calculate the tangential speed of the ball.



Q6

Analyze the Conical Pendulum further. Imagine you want the same toy plane to fly in a **WIDER circle** (meaning the angle θ becomes larger and the plane is higher up).

- (a) Even if the circle is wider, the plane is still making a horizontal circle without falling. Does the vertical upward force (T_y) need to change? Explain your reason. (Hint: Look at Gravity).
- (b) Because the angle θ is larger, the string is more "flat". To keep the SAME vertical force T_y while pulling at a flatter angle, must the total tension (T) in the string increase or decrease? Explain your reason.
- (c) To maintain this new, wider circular path, does the plane need to fly faster or slower? Explain your reason.

